

Learning Rate

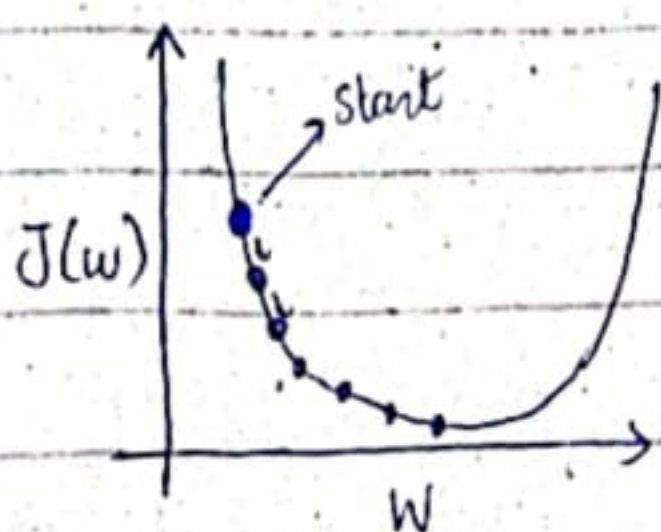
In this we will look at how choosing value of alpha affects the algorithm.

as :

$$W = W - \alpha \frac{dJ(W)}{dW}$$

lets look at what happens if the learning rate alpha is too small or if it's too large.

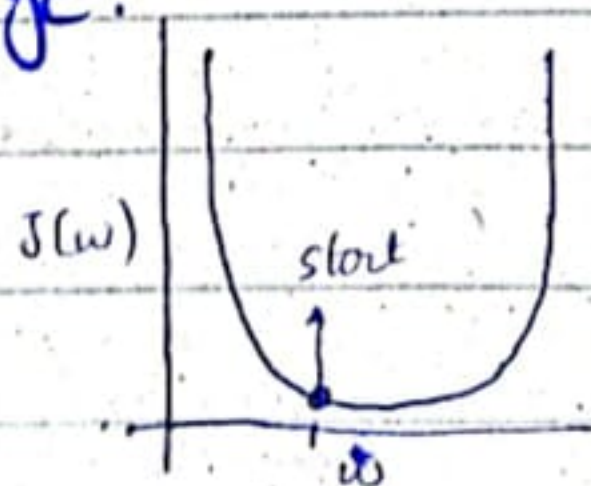
If α too small :



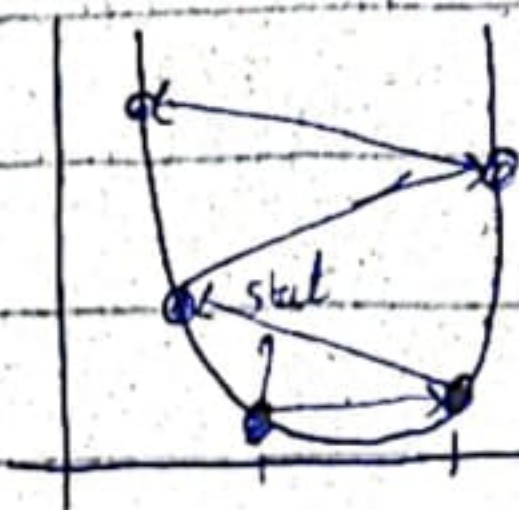
as the derivative multiplies by alpha.

when you multiply, i.e. $\alpha = 0.000001$, you end up with very small no. & like end up taking baby steps. You do end up close to min J but very slow. So, it will work but will be very slow.

If α too large :



As you see its very close to min J :



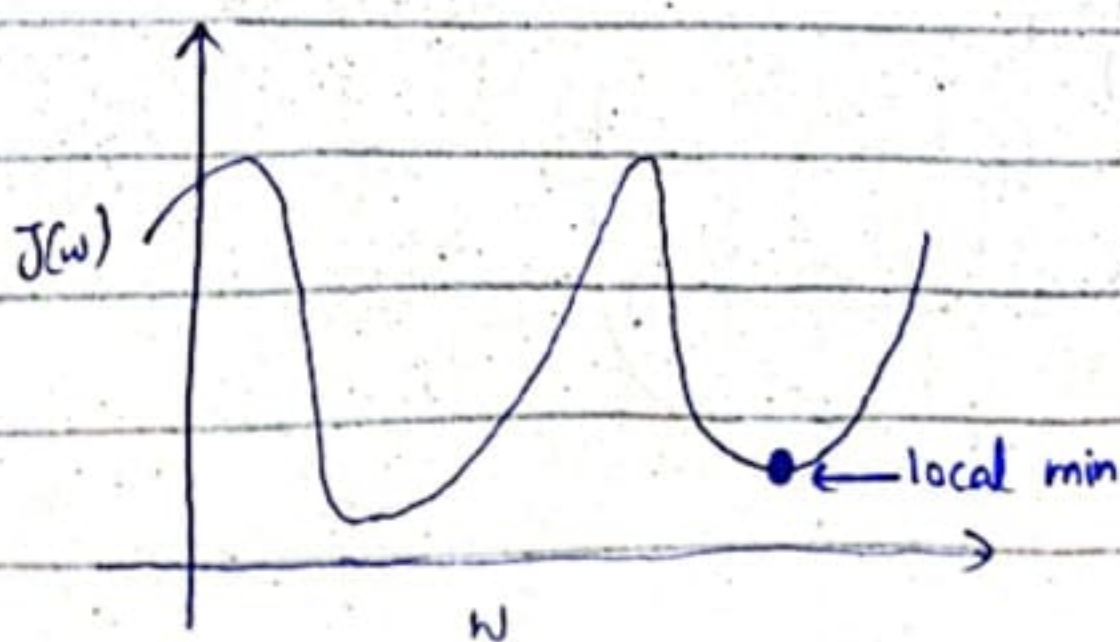
if α is too large
you will end up taking
a very large step
& skip the minimum
point.

- Now the cost ended up increasing & actually making it worse.
- The derivative now say to decrease w so you end up again taking large step & again worsening it as seen in graph.
- So you notice you are going further & further away from the minimum.

Gradient descent may:

- never reach min, overshoot
- Fail to converge, diverge

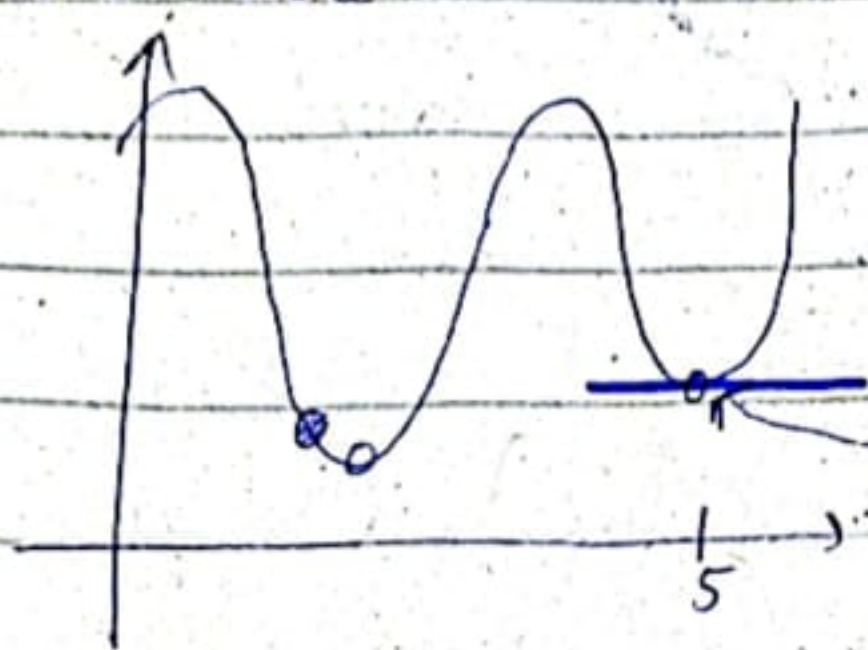
Another question you may wonder:



what if your w
is already at
this point here?
so cost J is
already at
local min

What you think one step of g.d will do if you already reached minimum?

Let suppose you have cost function J and the one you see in graph isn't sq. error c.f E . This has two local minima corresponding to two valleys you see there.



Suppose after some g.d steps your param w is here at 5 so you are on this point on c.f J .

When you check slope of this ^{current val of w} function is zero so derivative becomes 0.

So it becomes,

$$w = w - \alpha \cdot 0$$

$$\text{as } \frac{dJ(w)}{dw} = 0$$

$$\boxed{w = w}$$

So if you are at local minima gradient descent leaves w unchanged.

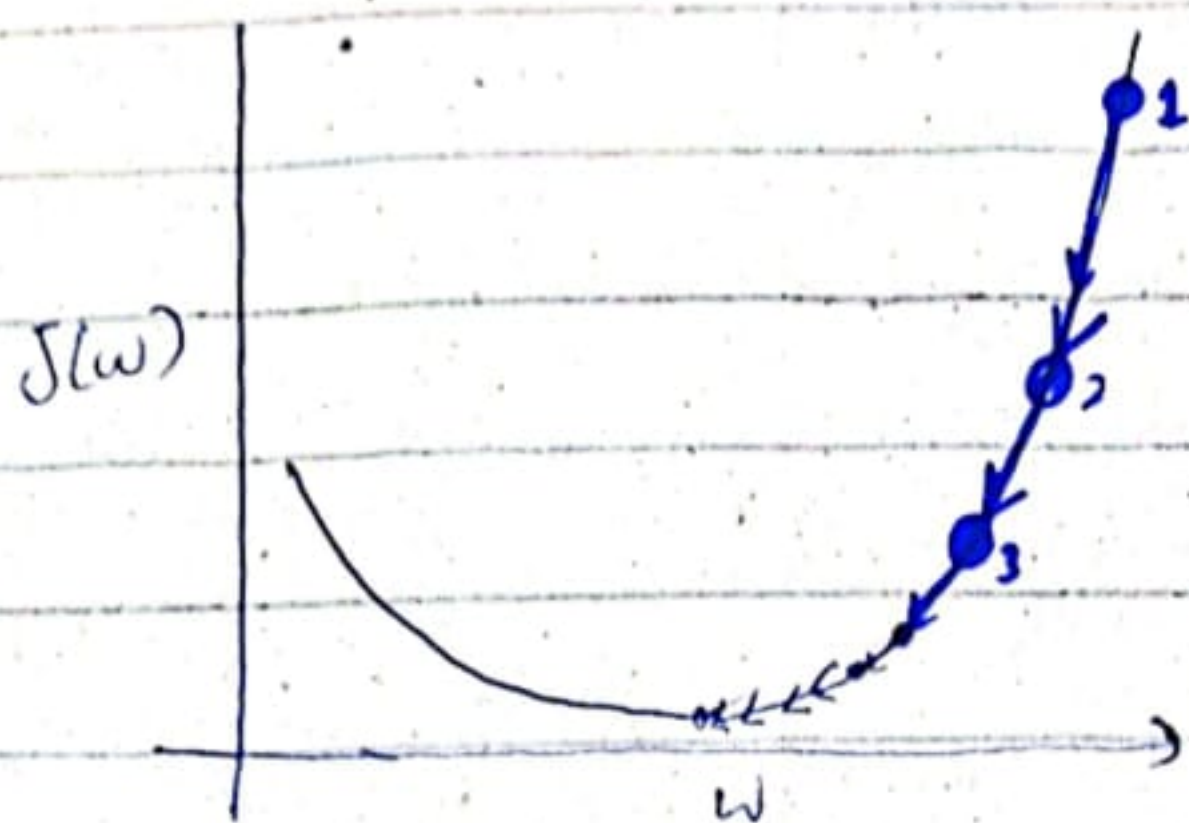
Now f.e take $w = 5$:

$$\text{So } w = 5 - 0.1 \cdot 0$$

$$\boxed{w = 5}, \quad 5 = 5$$

So if you reach local minima further gradient descent absolutely does nothing.

Can reach local minimum with fixed learning rate



- At "1" as slope is too steep, derivative becomes larger so $(a \times \text{derivative})$ is larger takes a big step.

- At "2" slope is less steep ^{derivative less} than when we were at 1 so step becomes a bit small.
 $a \times \text{derivates} \rightarrow$ not as large.

- At "3" slope becomes more smaller (less steep so derivative is small)

$a \times \text{derivative} \rightarrow \text{small} = \text{small step}$

→ so derivative at each step gets closer & closer to "zero", so in g.d we eventually reach small ϵ small steps until we reach minimum.

So we can use this to minimize any cost function not just sq. error regression ones.